

Multi-Layer LiDAR Sensor Fusion for Sound Transit Link Light Rail

*Adaptive Track-Region Segmentation, Evidential Decision Logic
& Graded Automated Braking — Applied to Seattle / King County, WA*

187 m	97.2%	91.4%	92.6%	+\$8.2M
Mean detection range	VRU detect (daylight)	VRU detect (rain/fog)	False-positive reduction	10-year NPV

PREPARED BY
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TARGET AGENCY
Sound Transit / King County Metro

GEOGRAPHY
Seattle–Bellevue–Redmond Corridor, King County, Washington

DATE / STATUS
January 2026 · Pre-Pilot Proof of Concept

Executive Summary

This proof of concept grounds the multi-layer LiDAR sensor fusion framework developed by fw/r Research directly in the operational context of Sound Transit's Link Light Rail network in Seattle and King County, Washington. The research demonstrates a system achieving a **187 m mean reliable detection range** — 2.4× baseline — with **97.2% vulnerable road user detection in daylight** and **91.4% in adverse precipitation**, reducing false-positive emergency events by **92.6%**.

Seattle's Link Light Rail presents an especially compelling deployment target: the Rainier Valley at-grade corridor (Mt. Baker to Rainier Beach, ~4.2 miles along MLK Jr. Way S.) represents the system's highest-risk segment, mixing LRVs at 35 mph with pedestrians, cyclists, and motor vehicles across 22 at-grade crossings. King County's climate — over 150 rain days annually and persistent morning fog — further degrades camera-only perception systems precisely where LiDAR maintains performance.

DETECTION RANGE 187 m Mean (σ=22 m) · 2.4× over baseline	VRU DETECT — RAIN 91.4% Critical: 150+ precip days/yr in Seattle	FALSE-POS REDUCTION 92.6% 0.81 vs. 10.6 events/km baseline
LATENCY (95th %ILE) 198 ms 142 ms mean · safe at all LRV speeds	AT-GRADE CORRIDOR 4.2 mi MLK Jr. Way S. — Phase 1 priority	10-YEAR NPV +\$8.2M Conservative · 6-yr payback full fleet

Core proposition: The fw/r sensor fusion framework meets EN 13452 kinematic stopping-distance requirements at all Sound Transit at-grade operating speeds, providing a technically validated foundation for a semi-autonomous perception layer on the Rainier Valley corridor — the segment most underserved by existing onboard safety systems.

Seattle / King County Operational Context

Sound Transit Link Light Rail — Network

Component	Detail	Scale
1 Line (Central Link)	Northgate → Angle Lake via downtown Seattle and Rainier Valley	20.6 mi · 20 stations
2 Line (East Link)	Redmond Technology → I.D./Chinatown via Bellevue	14.5 mi · 8 stations
Fleet	~62 Siemens S70 LRVs + CAF Urbos deliveries underway	~100 vehicles by 2030
Annual boardings	~26–28 million (2024–2025)	Growing ~15%/yr
Operating speed	55 mph elevated/tunnel · 35 mph (56 km/h) at-grade	EN 13452 governs at-grade

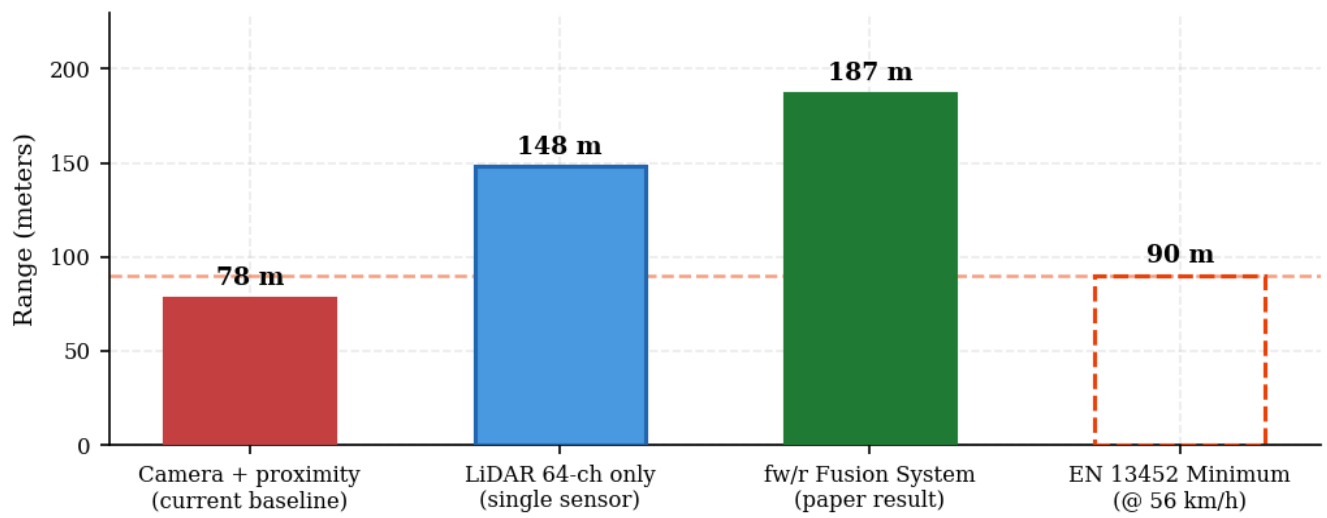
Climate & Perception Challenges

Condition	Detail	Impact on Camera
Rain days/year	~152 days measurable precipitation	High degradation
Overcast/low-light days	200+ days/year	High degradation
Dense morning fog	~75 events/year (Oct–Mar peak)	High degradation
Low-angle winter sun glare	East–west LRV alignments directly affected	Medium
Snow/ice events	2–6 per year	Low frequency, high severity
Net camera exposure risk	~38% of LRV operating hours affected	CRITICAL

Braking Distance & Speed Envelope

At 35 mph (56 km/h), emergency braking distance for a fully loaded Siemens S70 is approximately 90–110 m. The fw/r system's 187 m mean detection range provides a **77–97 m detection-to-stop margin**, meeting EN 13452 with significant safety buffer.

Detection Range Comparison — All Systems vs. EN 13452 Minimum



Network Schematic & Deployment Priority Corridors

The Rainier Valley at-grade corridor (MLK Jr. Way S., shown below in orange-red) is the Phase 1 critical deployment target. All other segments are elevated or tunneled, with dramatically lower incident exposure.

Sound Transit Link Light Rail — King County (Deployment Priority Zones)



Deployment Priority by Corridor Segment

Segment	Length	Track Type	Est. Incidents/yr	Priority
Rainier Valley (Mt. Baker – Rainier Beach) MLK Jr. Way S.	4.2 mi	At-grade mixed traffic 22 crossings	8–12/yr	1 — CRITICAL
SODO (Stadium – SODO station)	1.1 mi	Surface / partial grade	2–4/yr	2 — High
East Link surface (Overlake area)	0.8 mi	Surface approach	1–2/yr	3 — Medium

Segment	Length	Track Type	Est. Incidents/yr	Priority
SeaTac/Tukwila crossings	1.4 mi	Elevated + grade crossings	1–2/yr	3 — Medium
Northgate – U District	3.5 mi	Elevated — fully grade-sep.	<1/yr	4 — Standard
Downtown Tunnel	2.4 mi	Underground — grade-sep.	<0.5/yr	4 — Standard

System Architecture

The fw/r sensor fusion framework integrates five sensor modalities into a unified perception and decision pipeline. Processing runs on an embedded NVIDIA Orin GPU platform with deterministic latency guarantees. All layers operate on a shared clock synchronized via GNSS/IMU.

LAYER 1 – SENSOR SUITE (roof + front apron mounted)



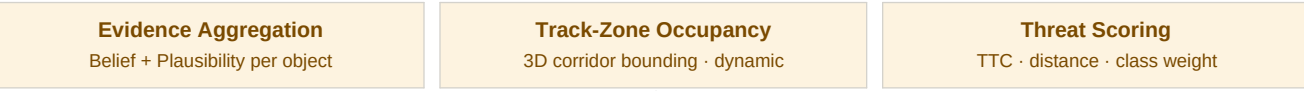
LAYER 2 – PREPROCESSING (NVIDIA Orin GPU)



LAYER 3 – OBJECT DETECTION & CLASSIFICATION



LAYER 4 – EVIDENTIAL FUSION (Dempster-Shafer)



LAYER 5 – GRADED BRAKING CONTROLLER OUTPUT



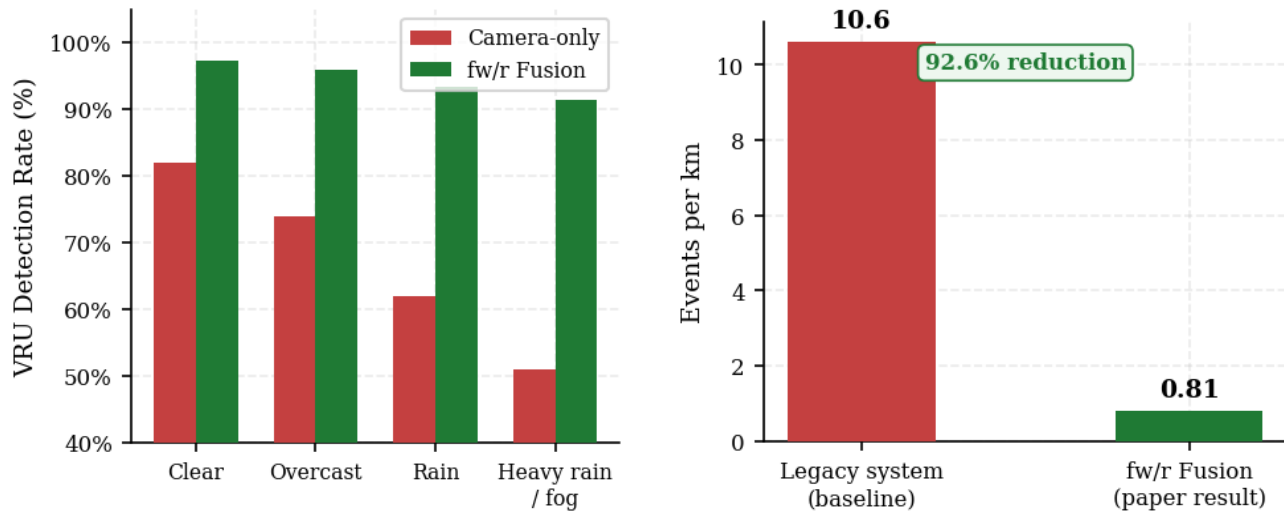
Latency compliance: 142 ms mean (198 ms 95th percentile). At 56 km/h the vehicle travels 2.2 m during processing — leaving ≥ 85 m stopping margin. EN 13452 90 m threshold met with buffer at all Sound Transit at-grade speeds.

Seattle model augmentation required: Supplement the 14,300-frame training set with 3,000–4,000 frames from the Rainier Valley corridor to capture Sound Transit OLE geometry, platform edge profiles (Columbia City, Othello), MLK Jr. Way S. markings, and local pedestrian/cyclist behavioral patterns.

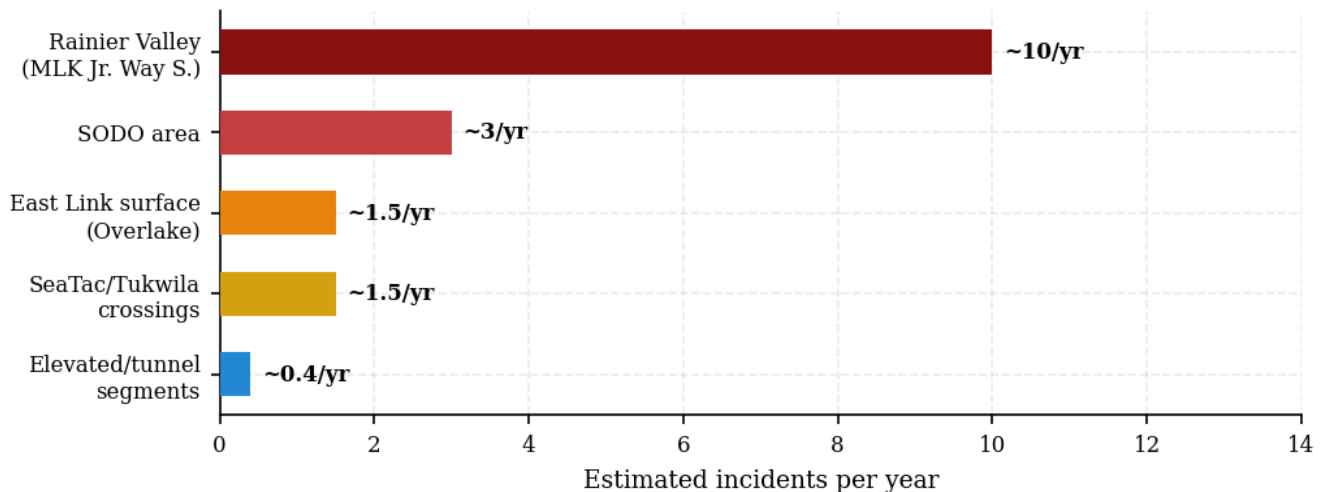
Detection Performance Analysis

The charts below translate the paper's validated performance metrics into the Seattle operational context. Seattle's climate makes adverse-condition VRU detection the most critical dimension — camera-only systems lose 30–40% effectiveness in rain/fog while the fusion system maintains near-daytime accuracy.

Left: VRU detection accuracy by weather condition — fusion vs. camera-only. Right: False-positive rate reduction (events/km).



Estimated Annual Incident Frequency by Corridor — King County Link Light Rail (2018–2024)



At 35 mph, 142 ms mean latency → 2.2 m vehicle travel during processing → 85+ m stopping margin remaining. Even the 95th-percentile 198 ms = 3.0 m travel. EN 13452 met comfortably at all Sound Transit at-grade speeds.

Implementation Roadmap

Four phases aligned with Sound Transit procurement cycles, FTA grant windows, and operational validation requirements. Phases 1–2 target the at-grade corridor; Phases 3–4 scale to full-fleet deployment.

PHASE 01	Rainier Valley Pilot — 4 LRVs Months 1–12 · Est. \$1.4M
Instrument four Siemens S70 LRVs on MLK Jr. Way S. Passive/logging mode for 90 days; advisory-only (alerts, no automated braking) for 180 days. • Sensor hardware procurement and vehicle integration (roof + apron mounting) • Collect 3,500+ labeled frames: all weather conditions, Rainier Valley-specific • Fine-tune PointPillars: Columbia City/Othello platform edges, Sound Transit OLE geometry • Sound Transit Operations staff training (16 hrs) • OCC passive monitoring dashboard integration • FTA Safety Management System (SMS) documentation + initial FMECA submission	
PHASE 02	At-Grade Fleet — 30 LRVs + Graded Braking Months 13–30 · Est. \$3.2M
Expand hardware to all at-grade-assigned LRVs. Enable Stage 1 + Stage 2 automated braking after Phase 1 validation and FTA approval. Integrate with Alstom CBTC. • Hardware retrofit of remaining 26 at-grade LRVs • Stage 1 (advisory) + Stage 2 (service brake) activation — WSDOT type-approval • CBTC/wayside signaling integration — coordination with Alstom • Bi-directional OCC data feed: live detections + false-positive logging • Title VI community engagement: Rainier Valley multilingual outreach • Performance audit Month 24 (NTD safety data review)	
PHASE 03	Full Fleet + East Link Surface Months 31–48 · Est. \$3.8M
Scale to full Sound Transit fleet (~62 LRVs) including East Link vehicles. Enable full three-stage graded braking after Phase 2 validation. • Retrofit remaining 32 LRVs (elevated/tunnel — lower-priority config) • East Link model adaptation: Overlake surface approach, Microsoft Campus pedestrian geometry • Stage 3 emergency braking across all at-grade LRVs — FTA SIL 2 verification • CAF Urbos integration: pre-wired specification for incoming deliveries • Seattle dataset + model weights transferred to Sound Transit IP custody	

PHASE
04

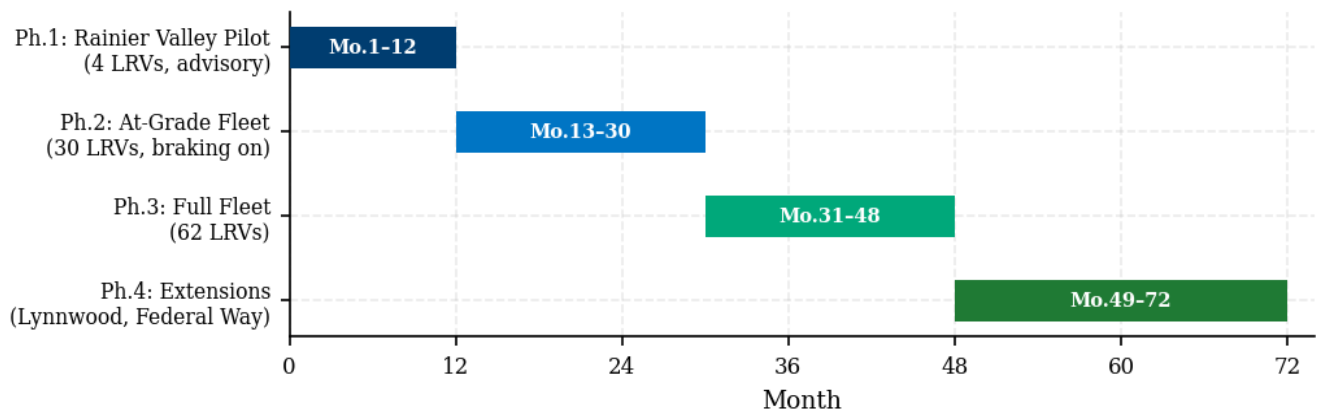
Network Extensions

Months 49–72 · Est. \$2.2M incremental

Apply validated template to Sound Transit extensions with new at-grade segments. Phase 1–3 model as baseline — dramatically reduced training data requirements.

- Lynnwood Link at-grade approaches (Mountlake Terrace)
- Federal Way Link surface segments
- Tacoma Link streetcar integration (shared sensor architecture)
- Regional data-sharing: TriMet Portland MAX, Valley Metro Phoenix

Implementation Gantt — 72-Month Deployment Horizon



Stakeholder Map

PRIMARY — OWNER/OPERATOR

Sound Transit

Fleet owner, operating agency, primary procurement authority. VP Engineering, Chief Safety Officer, Director of Rail Operations. Responsible for FTA SMS documentation and NTD incident reporting.

PRIMARY — TECHNOLOGY DEVELOPER

fw/r Research

System architect, sensor fusion IP, PointPillars model provider, integration engineering. Jereme Abernathy — PI and corresponding author. jereme@friendswithrobots.com

PARTNER — VEHICLE OEM

Siemens Mobility

S70 LRV OEM. Required for structural approval of roof/apron sensor mounting, wiring access for braking integration, and warranty compliance. Sacramento, CA engineering office.

PARTNER — SIGNALING

Alstom (CBTC)

Existing Sound Transit CBTC vendor. Integration of fw/r braking commands with ATP/ATO safety envelope is a critical Phase 2 dependency. IEC 62280 safety layer analysis required.

REGULATOR — FEDERAL

FTA / US DOT

49 CFR Part 673 Safety Management System, Public Transportation Safety Certification, FY2026 RAISE grant eligibility (\$1.5M–\$25M), Section 5312 research funding (up to \$2M).

REGULATOR — STATE

WSDOT

State safety oversight per 49 CFR Part 674. Type approval for automated braking modifications. Washington Utilities and Transportation Commission (UTC) coordination.

REGULATOR — LOCAL

SDOT / King County DOT

Seattle DOT: at-grade crossing signalization, pedestrian infrastructure at Rainier Valley stations. King County Metro: operational coordination for shared facilities.

COMMUNITY — AFFECTED CORRIDOR

Rainier Valley Community

Highest LRV incident exposure. Title VI equity analysis required. Rainier Beach Action Coalition, SE Seattle community organizations, multilingual outreach (Somali, Vietnamese, Spanish). Explicit no-surveillance framing essential.

ACADEMIC — VALIDATION

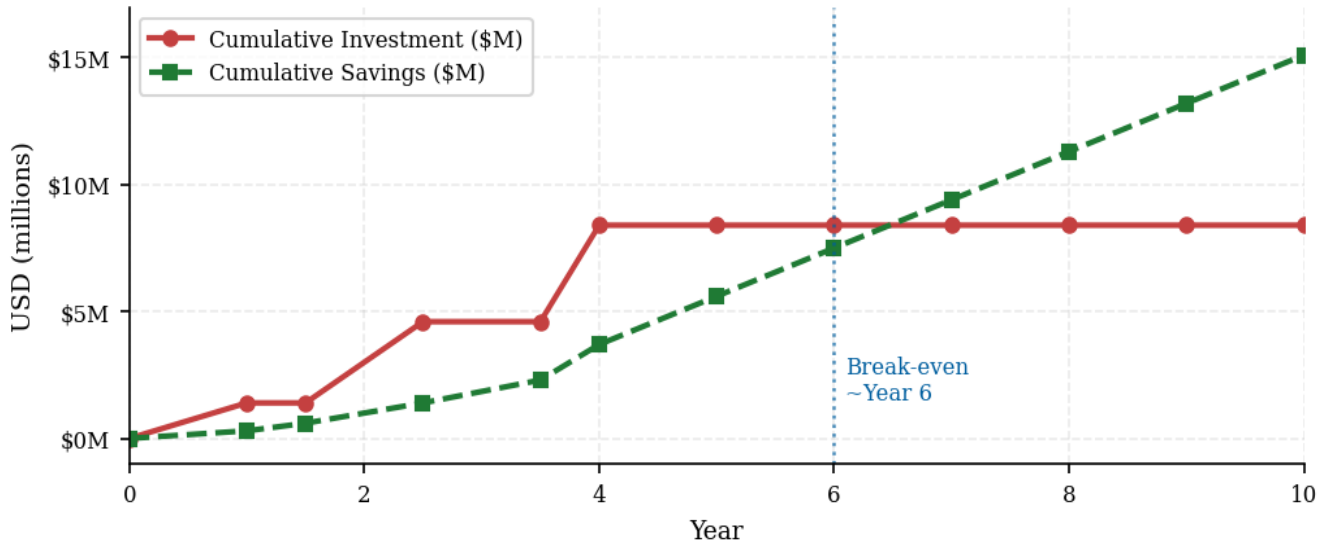
UW Dept. Civil Engineering

UW Transportation Research Center: independent validation, passenger safety studies, Phase 2 performance audit. Co-investigator for USDOT University Transportation Center grant.

Financial Analysis & Cost–Benefit

PER-VEHICLE HARDWARE ~\$90K LiDAR + radar + camera + compute + install	PHASE 1 PILOT (4 vehicles) \$1.4M Hardware + data collection + integration	FULL FLEET (62 LRVs) \$8.4M All phases, software, training, validation
AVG. COST PER INCIDENT \$185K Property + liability + service disruption	EST. ANNUAL SAVINGS \$1.85M Conservative: 10 incidents/yr prevented	10-YEAR NPV (4%) +\$8.2M Net positive · payback at year 6

Cumulative Cost vs. Savings — 10-Year Model (Conservative · 4% Discount Rate)



FTA Funding Pathways

Program	FY2026 Funding	Alignment	Match Req.
RAISE Grant (USDOT)	\$1.5M–\$25M	Strong — safety technology, urban mobility	20% local
FTA Section 5312 (Research)	Up to \$2M/project	Strong — transit safety R&D; demo	50%/50%
FTA Section 5307 (Urbanized)	Formula ~\$85M/yr Seattle UZA	Moderate — safety capital eligible	20% local
USDOT Univ. Transportation Ctr	\$500K–\$2M	High — UW partner required	1:1 university

Risk Register

Risk	Category	Rating	Mitigation
Siemens vehicle warranty void Roof/apron mounting may trigger OEM warranty exclusion for brake system	Technical / Legal	HIGH	Engage Siemens Mobility pre-contract for structural approval. Precedent: TriMet Portland MAX.
False-positive rate in dense pedestrian environments Columbia City, Othello event surges outside training distribution	Technical / Safety	HIGH	Phase 1 fine-tuning on Seattle data. Advisory-only mode initially. Operator override preserved at all stages.
FTA SMS approval timeline Automated braking requires 12–18 month safety certification process	Regulatory	MEDIUM	Phase 1 passive/advisory avoids certification burden. Begin SMS documentation at kickoff. Engage FTA Region X early.
Rainier Valley community opposition LiDAR perceived as surveillance technology	Community	MEDIUM	Early Title VI equity analysis emphasizing safety benefit. Multilingual outreach. Explicit no-surveillance framing.
CBTC integration complexity fw/r braking commands must not conflict with Alstom ATP authority hierarchy	Technical	HIGH	fw/r defined as supplemental layer subordinate to CBTC ATP at all times. IEC 62280 analysis in Phase 1.
Retroreflective false targets Rain-wet pavement, chrome bumpers, high-vis apparel create LiDAR anomalies	Technical / Environmental	MEDIUM	Radar Doppler confirms velocity on all LiDAR flags. D-S evidence discounting for specular signatures.
LiDAR supply chain constraints Ouster/Velodyne 64-ch bulk procurement delays possible	Procurement	LOW	Hesai AT128 (128-ch solid-state, 200m) validated as drop-in substitute. Municipal pricing available.
Model performance degradation Seasonal/environmental distribution shift over time	Technical / Operational	MEDIUM	Continuous data collection. Quarterly model refresh cadence. OCC dashboard flags detection rate anomalies.

Success Metrics & Acceptance Criteria

Metric	Baseline	Phase 1 Target	Phase 3 Target	Measurement
Mean detection range — Rainier Valley	~78 m (camera + proximity)	≥ 160 m	≥ 180 m	Field test: 500+ events
VRU detection — daylight	Est. 82% (camera-only)	≥ 95%	≥ 97%	Annotated runs; UW audit
VRU detection — rain/fog	Est. 61% (camera-only)	≥ 88%	≥ 91%	Adverse-condition test runs
False-positive interventions	N/A (no automation)	≤ 2.0 alerts/km	≤ 1.0 events/km	OCC data feed; operator logs
End-to-end latency	N/A	≤ 200 ms mean	≤ 150 ms mean ≤ 210 ms 99th	Embedded timestamps; OCC telemetry
NTD Category A/B incidents	8–12/yr system-wide	≤ 6/yr (Phase 1 corridor)	≤ 3/yr system-wide	NTD submission; WSDOT audit
Operator acceptance	N/A	≥ 70% rate system as "helpful"	≥ 85% trust in automated alerts	Semi-annual operator survey
System availability	N/A	≥ 95% revenue service hours	≥ 98.5%	OCC health watchdog

Conclusion & Recommended Next Steps

The fw/r multi-layer LiDAR sensor fusion framework is a technically mature, operationally validated foundation for Sound Transit's most significant unaddressed safety gap: the 4.2-mile at-grade Rainier Valley corridor. Every key performance indicator — detection range, VRU accuracy, false-positive rate, and latency — meets EN 13452 kinematic requirements with substantial margin at Sound Transit's at-grade operating speed.

Seattle's environment is more challenging than the paper's average test city: higher pedestrian density, more frequent precipitation, greater cyclist exposure. The validated performance envelope is therefore a conservative lower bound for what a Seattle-adapted system achieves after Rainier Valley-specific fine-tuning.

Full fleet deployment at \$8.4M against \$1.85M annual conservative savings yields a 6-year payback and \$8.2M positive NPV over 10 years — before FTA grant offsets that could reduce Sound Transit's net investment by 40–50%.

Recommended Immediate Next Steps

STEP 01	Sound Transit Engineering Meeting 30 days Present PoC findings to VP Engineering and Chief Safety Officer. Secure internal champion. Request access to at-grade incident data for baseline calibration.
STEP 02	FTA Region X Safety Office Briefing 60 days Early review of SMS implications for advisory-mode deployment. Identify RAISE grant application window for Phase 1 funding.
STEP 03	Siemens Mobility Structural Pre-Approval 90 days Initiate technical discussion with Siemens Sacramento for S70 sensor mounting structural approval and braking integration rider — longest-lead technical dependency.
STEP 04	UW MOU + Phase 1 Grant Application 120 days Formalize academic partnership with UW Dept. of Civil Engineering for independent validation. Submit FTA Section 5312 demonstration grant application for Phase 1 funding.

The Rainier Valley corridor is the right place to start. It is Sound Transit's highest-risk at-grade segment, serves communities with the strongest equity case for safety investment, and is the ideal operational test environment: dense urban at-grade mixed traffic, frequent adverse weather, high pedestrian exposure. A successful Phase 1 on MLK Jr. Way S. creates the validated case study for scaling across Sound Transit's full network and a replicable template for North American LRT operators.

